Lugiato-Lefever Equation: Time-evolution Solver

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1 Introduction

Lugiato-Lefever equation, shortly known as the LLE, is a spatiotemporal model that can be used to characterize Kerr-comb generation in microresonators. In this model, the evolution is in time, and the conjugate (transverse) variable is the resonator's azimuthal angle. It can be considered as a variant of the nonlinear Schrödinger equation (NLSE), and it includes damping, driving, and detuning.

In its normalized form, LLE can be expressed as

$$\frac{\partial \psi}{\partial \tau} = -(1 + i\alpha)\psi + i\gamma \left|\psi\right|^2 \psi - i\frac{\beta_2}{2}\frac{\partial^2 \psi}{\partial \theta^2} + F, \tag{1}$$

where ψ is the normalized amplitude, τ is the normalized time, γ is the nonlinearity coefficient, β is the dispersion parameter, α is detuning, and F is the pump power. The general LLE is normalized with respect to gain and loss. Detailed information can be found in Ref. [1].

2 Numerical Solver Methodology

For the time-evolution solution, we solve it in a way that avoids being computationally expensive or inaccurate. A split-step Fourier method is used to solve the equation. In this method, one fractional step involves taking the Fourier transform to solve the corresponding ODE, while the other fractional step explicitly solves the ODE in the regular domain. This approach is used because some parts of the problem are easier to solve in the Fourier domain, while others are more manageable in the regular domain. In our case, the linear part is easier to solve in the Fourier domain, whereas the nonlinear part is more straightforward in the regular domain. Therefore, the linear and nonlinear parts of the LLE are solved sequentially, with the solution of one part serving as the initial condition for the other. The transformations are performed analytically to ensure computational efficiency.

Now, the commutative property does not hold this way, which means solving first the linear part does not match the solution when solving first the nonlinear part. And this introduces inaccuracy of the first order into the solution. To reduce the order of inaccuracy, the linear part is solved for half a step, then the nonlinear part is solved for a full step, and then the linear part again is solved for the rest of the half-step. This can also be done in reverse order. We can continue to split the steps to decrease order of inaccuracy, but the process becomes computationally expensive. That is why we keep the splitting to half-step. Please note that the solution of each sequence is used as the initial condition of the following sequence.

2.1 Linear Part

First, we consider the linear part. If we rewrite the LLE considering only the linear part, it looks like below:

$$\frac{\partial \psi}{\partial \tau} = \left[-(1 + i\alpha) - i\frac{\beta}{2}\frac{\partial^2}{\partial \theta^2}\right]\psi + F \tag{2}$$

In its current form, the above expression is not easy to solve, since the operator $\partial^2/\partial\theta^2$ will be a complicated matrix. Therefore, we take the Fourier transform of this equation from θ domain to μ (relative mode number) domain. The resulting expression looks like:

$$\frac{\partial \tilde{\psi}}{\partial \tau} = \left[-(1 + i\alpha) - i\beta(i\mu)^2\right]\tilde{\psi} + \tilde{F}(\mu) \tag{3}$$

Here, $\tilde{\psi}$ and $\tilde{F}(\mu)$ correspond to the electric field amplitude and pump amplitude in the mode number domain after the Fourier transform, and $\mathcal{F}(\partial^2/\partial\theta^2) = (i\mu)^2$. We let $\tilde{A} = -(1 + i\alpha) - i\beta(i\mu)^2$, and the equation becomes:

$$\frac{\partial\tilde{\psi}}{\partial\tau} = \tilde{A}\tilde{\psi} + \tilde{F}(\mu) \tag{4}$$

Eq. 4 is an ODE that can be solved using separation of variables. The ODE can be rewritten as,

$$\frac{1}{\tilde{A}\tilde{\psi} + \tilde{F}}\partial\tilde{\psi} = d\tau$$

In order to perform integration of the LHS, let us assume, $\tilde{A}\tilde{\psi} + \tilde{F} = v$, which implies $\partial\tilde{\psi} = \partial v/\tilde{A}$. Replacing the assumption in the ODE, we obtain,

$$\begin{aligned}
\frac{1}{\tilde{A}}\int\frac{1}{v}dv &= \int dt \\
\Rightarrow \ln(\tilde{A}\tilde{\psi} + \tilde{F}) &= \tilde{A}t + C_0 \\
\Rightarrow \tilde{A}\tilde{\psi} + \tilde{F} &= e^{\tilde{A}t + C_0} \\
\Rightarrow \tilde{\psi} &= \frac{1}{\tilde{A}}e^{\tilde{A}t + C_0} - \frac{\tilde{F}}{\tilde{A}} \\
\Rightarrow \tilde{\psi} &= \frac{e^{C_0}}{\tilde{A}}e^{\tilde{A}t} - \frac{\tilde{F}}{\tilde{A}}
\end{aligned}$$

Therefore, assuming $C_1 = \frac{e^{C_0}}{\tilde{A}}$, general solution to the differential equation posed in Eq. 4 is:

$$\tilde{\psi}(\tau) = C_1 e^{\tilde{A}\tau} - \frac{\tilde{F}}{\tilde{A}}, \tag{5}$$

where, C_1 is a constant. In order to find C_1, we imply initial conditions. At $\tau = \tau_0$,

$$\begin{aligned}
\tilde{\psi}(\tau_0) &= C_1 e^{\tilde{A}\tau_0} - \frac{\tilde{F}}{\tilde{A}} \\
\Rightarrow C_1 e^{\tilde{A}\tau_0} &= \tilde{\psi}(\tau_0) + \frac{\tilde{F}}{\tilde{A}} \\
\Rightarrow C_1 &= \left[\tilde{\psi}(\tau_0) + \frac{\tilde{F}}{\tilde{A}}\right] e^{-\tilde{A}\tau_0}
\end{aligned} \tag{6}$$

Inserting Eq. 6 into 5, we obtain,

$$\tilde{\psi}(\tau) = \left[\tilde{\psi}(\tau_0) + \frac{\tilde{F}}{\tilde{A}}\right] e^{(\tau-\tau_0)\tilde{A}} - \frac{\tilde{F}}{\tilde{A}}, \tag{7}$$

In our last step, we take inverse Fourier transform of Eq. 7 to find $\psi(\tau)$.

2.2 Nonlinear Part

Since we have obtained solution for the linear part, we move towards solving the nonlinear part of the system. Let us rewrite the nonlinear part first,

$$\frac{\partial\psi}{\partial\tau} = i\gamma|\psi|^2\psi \tag{8}$$

If we assume that change in ψ is very small, therefore, $|\psi|^2$ does not change much with time ($|\psi(\tau)| = |\psi_0|$), and the nonlinearity coefficient also remains constant, Eq. 8 can be solved using simple separation of variables method:

$$
\begin{aligned}
\frac{\partial\psi}{\psi} &= i\gamma|\psi_0|^2\partial\tau \\
\Rightarrow \int \frac{\partial\psi}{\psi} &= i\int \gamma|\psi_0|^2\partial\tau \\
\Rightarrow \ln[\psi(\tau)] &= i[\gamma|\psi_0|^2\tau] + C \\
\Rightarrow \psi(\tau) &= C_2 e^{i[\gamma|\psi_0|^2\tau]}
\end{aligned}
\tag{9}
$$

Applying initial conditions as at $\tau = \tau_0$, $\psi = \psi_0$, we get,

$$
\begin{aligned}
\psi_0 &= C_2 e^{i[\gamma|\psi_0|^2\tau_0]} \\
\Rightarrow C_2 &= \psi_0 e^{-i[\gamma|\psi_0|^2\tau_0]}
\end{aligned}
\tag{10}
$$

Putting C_2 from Eq. 10 into Eq. 9, we get,

$$
\psi(\tau) = \psi_0 e^{i[\gamma(|\psi_0|^2\tau - |\psi_0|^2\tau_0]} \tag{11}
$$

Of course, it is clear now that the ψ_0 comes from the solution of the linear part.

3 Implementation

So far, we have obtained the two following formulae for linear and non-linear parts respectively:

$$
\psi(\tau) = \text{IFFT}\left\{\left[\tilde{\psi}(\tau_0) + \frac{\tilde{F}}{\tilde{A}}\right] e^{\tilde{A}(\tau-\tau_0)} - \frac{\tilde{F}}{\tilde{A}}\right\} \tag{12}
$$

$$
\psi(\tau) = \psi_0 e^{i[\gamma(|\psi|^2\tau - |\psi_0|^2\tau_0)]} \tag{13}
$$

where, $\tilde{A} = -(1 + i\alpha) - i\beta(i\mu)^2$. Overall, we have learned that the solution of Eq. 12 should be used as the initial condition for Eq. 13. Also, the linear part needs to be solved for half-step first with an assumed initial condition. Then the other half-step should be solved after the nonlinear part.

To do that, we discretize our timeframe with a small step size, let us say that is $\Delta t = h$. Then with the assumed initial condition, we solve for the linear part with $\Delta t = h/2$. Let us suppose the solution from the linear part is ψ_l. Hence, the initial condition for nonlinear part will be $\psi_0 = \psi_l$ and the time step should be $\Delta t = h$. Finally, if the solution from the nonlinear part is ψ_{nl}, then the initial condition for the other half-step of linear part will be $\psi_0 = \psi_{\text{nl}}$ and the time step will be $\Delta t = h/2$.

4 Sample Code

A basic code to implement the LLE time-evolution in `Matlab` that considers only second-order dispersion and one pump can be found in this GitHub repository. A basic simulation is done with this simple code and the solution output is presented in Fig. 1.

So far we have considered microresonators that only have second order dispersion. Now, what will change if our system has higher order dispersion? How do we include 3rd, 4th or even 5th orders in our LLE and how do we solve the LLE numerically?

References

[1] Yanne K. Chembo and Curtis R. Menyuk. "Spatiotemporal Model for Kerr Comb Generation in Whispering Gallery Mode Resonators". In: *Physical Review A* 87 (5 2013), p. 053852. DOI: 10.1103/PhysRevA.87.053852. URL: https://link.aps.org/doi/10.1103/PhysRevA.87.053852.

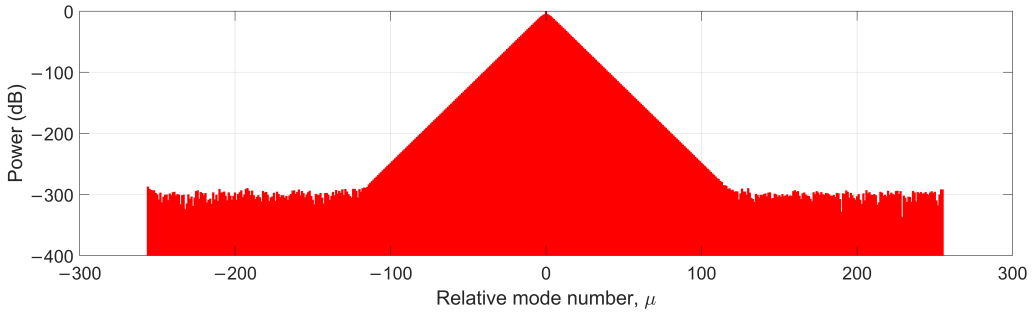


Figure 1: Single soliton solution after 100 photon lifetime evolution obtained from the basic code described in this report.